Report : A5 - A Machine Learning Competition
LINFO2262

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May 8, 2026

1 Protocol and model selection

To undertake this assignment, the initial step involved comparing the average performance and variability of several candidate models, ranging from simple logistic regression algorithms to deep neural networks. As illustrated in figure 1, the ExtraTrees classifier demonstrated the most robust performance among the six evaluated models.

The model was optimized using a two-stage hyperparameter tuning process via the automated optimization framework 'Optuna'. Indeed, rather than utilizing a single random search, a broad-to-focused optimization strategy was implemented. This consisted of a broad search spanning 50 trials to explore a wide parameter space, followed by a focused search of 30 trials designed to analyze more deeply the top 20% of the trials from the broad search. Also, `ccp_alpha` and `min_impurity_decrease` were included in the hyperparameter search to prevent overfitting.

2 Threshold Calibration and Cross-Validation

To maximize performance on the competition metric, the model incorporates threshold calibration based on the Balanced Classification Rate (BCR). By default, scikit-learn applies a probability threshold of 0.5. So, if the probability of belonging to the positive class is ≥ 0.5, the model predicts 1; otherwise, it predicts 0. However, given that the dataset is highly imbalanced (90% negative, 10% positive), the model tends to produce lower probabilities for the positive class. As a result, maintaining the default 0.5 threshold would result in predicting almost all instances as 'negative', significantly reducing the ability to identify true positives and lowering the overall BCR.

To address this issue, a `Calibrate_threshold` function was implemented to determine the decision threshold that maximizes BCR for given true labels and predicted positive probabilities. The optimal threshold was found to be 0.275 which means that the positive class needs a lower cutoff to optimize balanced accuracy.

Additionally, the model is evaluated with repeated stratified 5-fold cross-validation and the decision threshold is calibrated on the out-of-fold probabilities to maximize balanced accuracy. This step is essential to deal with our unbalanced dataset as it ensures that the class distribution in each fold reflects the global distribution and reduces the impact of stochastic variation in splits, thus providing a stable mean performance estimate.

3 Final Prediction Procedure

The prediction procedure for the final ExtraTrees model was generated through the following steps:

1. **Parameter Selection:** The best hyperparameters from the focused Optuna search were extracted (see table 1).

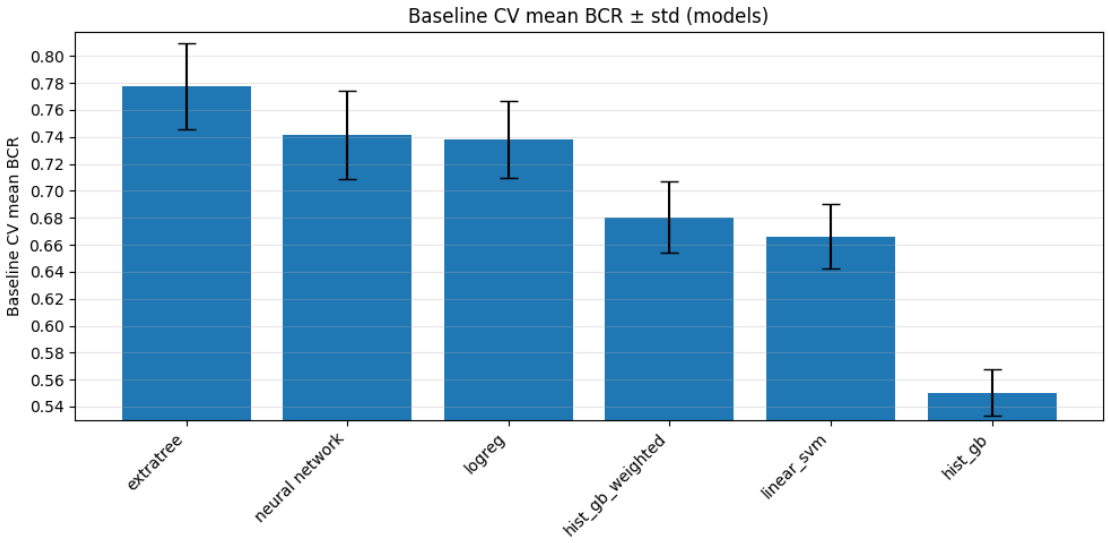


Figure 1: Mean balanced classification rate (BCR) ± 1 std across repeated CV for each candidate model.

2. **Full Model Training:** The model was retrained on the entire labeled training set using the optimized parameters.

3. **Threshold Application:** The optimal threshold identified during cross-validation was applied to the raw probabilities of the `x_test` dataset.

Parameter	**values**
`n_estimators`	2650
`ccp_alpha`	0.003
`class_weight`	balanced subsample
`max_features`	0.5
`bootstrap`	False
`max_depth`	20
`min_samples_split`	3
`min_samples_leaf`	10
`criterion`	entropy
`min_weight_fraction_leaf`	0.003
`min_impurity_decrease`	0.0016
`max_samples`	None

Table 1: Final model hyperparameter values.

As a result of those implementations, the final prediction split yielded 815 negative and 185 positive, confirming that the model does not collapse to a single majority class. The ExtraTrees model achieved a baseline CV mean BCR of 0.77759 with a CV standard deviation of 0.03228, which is both high and reasonably stable across the 5 CV folds.

However, because this CV mean BCR is measured on validation folds using a model trained strictly on training folds, it can occasionally overestimate performance on unseen test data. Therefore, a more conservative estimate was calculated for submission by subtracting 0.5 × standard error of the mean from the CV mean, resulting in an expected BCR of 0.77343. This conservative adjustment accounts for the likely gap between cross-validation performance and true test performance and hedges against overfitting the hyperparameters to the CV folds. One thing to notice is that the gap between the mean CV score

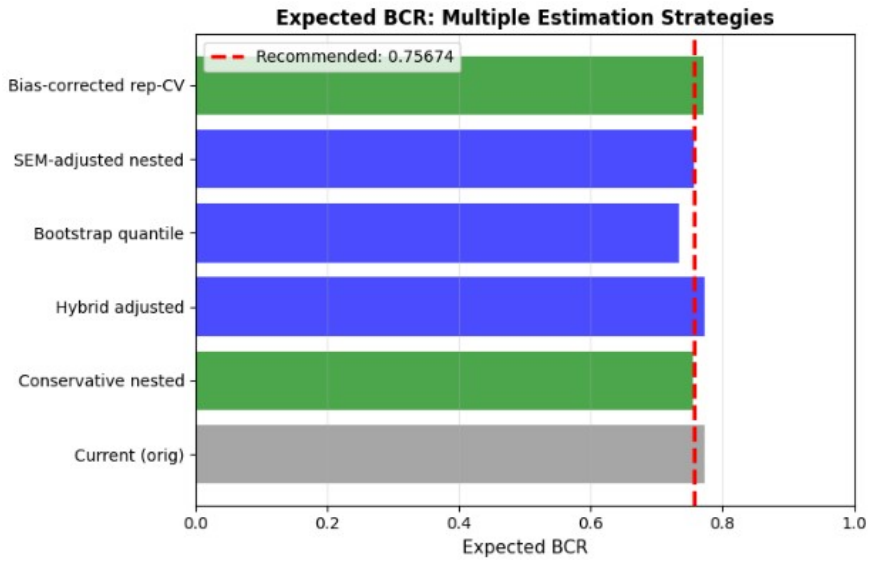


Figure 2: Comparison of expected BCR estimation strategies.

and the conservative submission estimate is quite small. So, the model is not only strong on CV but also fairly robust. To reduce optimism from model selection even further, the code also computes a nested cross-validation estimate and combines both protocol into a final refined expected BCR. This is done by taking the median value of the 3 different approaches depicted in blue in figure 2 which gives 0.75674. This final value is intended to be a more robust and defensible estimate of the model's likely performance on the hidden test set.

4 Feature Importance and Dimensionality

As shown in figure 3, a very small set of features (notably `x_192` and `x_352`) dominates the model's split-based importance, while the remaining features form a long tail of smaller contributions. These top twenty features represent the strongest predictors based on mean decrease in impurity.

In order to quantify the predictive power captured by these features, the model was retrained using only the top 5 and top 10 features, with the resulting validation BCR and AUC metrics displayed in table 2. The results indicate that the top 10 features preserve about 92% of the full model's balanced accuracy, while

Model	n_features	Threshold	BCR	AUC	BCR retention (%)	AUC retention (%)
Full model	1024	0.2750	0.7776	0.8320	100.0000	100.0000
Top 10 features	10	0.4700	0.7180	0.7542	92.3315	90.6519
Top 5 features	5	0.5550	0.6954	0.6925	89.4261	83.2295

Table 2: Repeated stratified cross validation results showing calibrated decision threshold, BCR, ROC AUC and BCR/AUC retention for top 10 and top 5 feature subsets relative to the full model.

shrinking the subset to the top 5 still preserves a lot of balanced accuracy but loses a more noticeable amount of ranking power, with the AUC dropping to 83% of the full model's value.

5 Evaluation and confusion matrix analysis

The aggregated out-of-fold confusion matrix (figure 4) reveals that the model is significantly more effective at recognizing negative instances than positive ones. Out of 2,700 actual negative cases, it correctly classifies

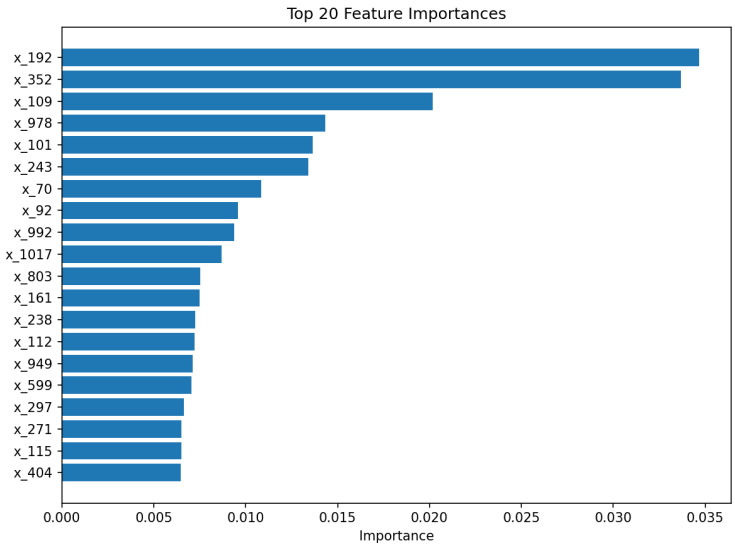


Figure 3: Horizontal bar chart of the 20 most important features (by mean impurity decrease) from my final ExtraTrees model.

2,372 and misclassifies 328 as positive, yielding a negative recall of about 88%. For the positive class, it correctly identifies 207 out of 300 cases, resulting in a positive recall of about 69%. Therefore, the primary limitation of the model lies in its precision for the positive class. Because many predicted positives are false alarms, the positive precision sits at approximately 39%. Overall, the model preserves strong performance on the majority class while deliberately trading off precision to capture more minority class positives.

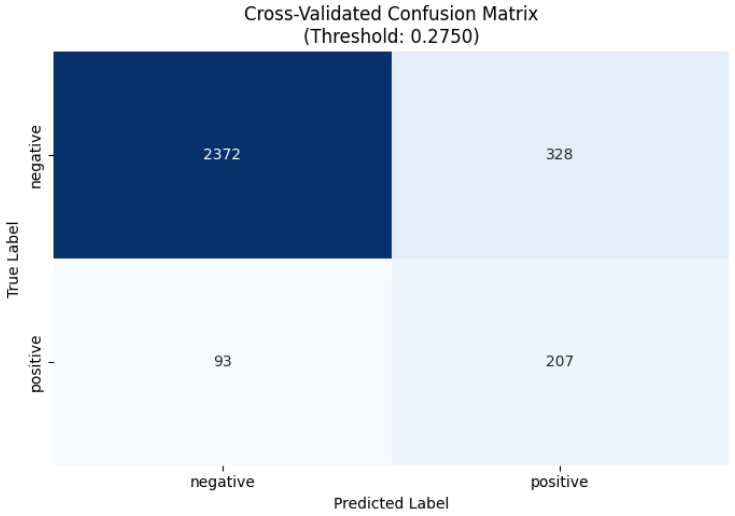


Figure 4: Aggregated out-of-fold confusion matrix at the calibrated decision threshold showing true vs predicted counts for negative/positive.

It is important to emphasize that the confusion matrix, ROC curve and threshold selection are derived exclusively from aggregated out-of-fold predictions, representing validation estimates rather than training set performance. Furthermore, because "RepeatedStratifiedKFold" produces overlapping validation folds, the standard `cross_val_predict` utility could not be utilized directly. To address this, a single calibrated prediction was generated for each training sample by averaging the probabilities produced for that specific

instance across every cross-validation fold in which it was held out. This approach ensures that the performance metrics reflected in figure 4 and the ROC analysis are realistic indicators of how the model generalizes to unseen data.

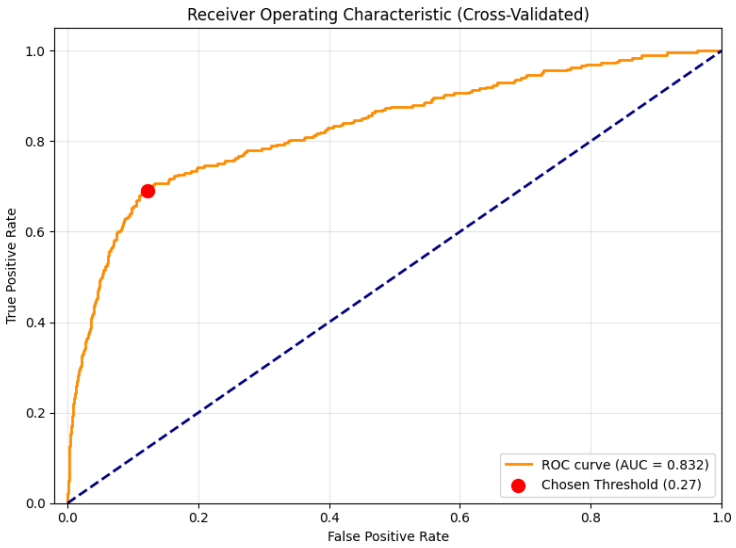


Figure 5: ROC curve computed from aggregated out of fold probabilities with AUC; includes a marker for the chosen operating threshold (0.275).

Finally, the ROC curve in figure 5 confirms the model's strong discriminative power. The orange curve stays well above the diagonal baseline and the AUC of 0.832 indicates a solid ability to separate positive from negative classes. The chosen threshold of 0.275 lies on the part of the curve where the model achieves a relatively high true positive rate of about 0.69 while keeping the false positive rate moderate, around 0.12. In practical terms, this threshold favors detecting more positive cases, at the cost of some extra false alarms, which is reasonable for our imbalanced classification problem where missing positives is more costly than predicting a few false positives.

6 Conclusion

In summary, through systematic evaluation, the ExtraTrees classifier emerged as the most robust architecture. By combining a two-stage hyperparameter optimization via Optuna with a customized probability threshold calibration, the model effectively mitigates the severe class imbalance, successfully prioritizing the detection of the minority positive class.

While the model inherently trades positive precision for higher recall thus resulting in a higher false positive rate, this is a calculated and necessary compromise to maximize the BCR. Furthermore, the feature importance analysis revealed a highly concentrated predictive signal, demonstrating that the model can retain over 92% of its balanced accuracy using only the top 10 features. Finally, with a conservative expected test BCR of 0.75674, the finalized model demonstrates strong discriminative power and stability across cross-validation folds, making it a competitive solution for this assignment.